

Section 3.6: Mutually Exclusive and Independent Events

Finite Mathematics MTH 107 Fall 2014

Dr. James Ashe

Mars Hill University

Monday 20th October, 2014

/home/jim/JamesAshe/MHU/images/MarsHillU

Independent/Dependent Events

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Independent

Two events, A and B , are *independent* if

$$p(A | B) = p(A).$$

Knowing B does not effect A 's probability.

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Two cards are dealt (without replacement). Are the events independent?

J: A jack is dealt 1st

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Example 3.6.1.

Two cards are dealt (without replacement). Are the events independent?

J: A jack is dealt 1st

H: A queen is dealt 2nd

No. Dependent.

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Example 3.6.2.

Two dice are rolled. Are the events independent?

4: The 1st die is 4

2: The 2nd die is 2

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Two dice are rolled. Are the events independent?

4: The 1st die is 4

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Yes. Independent.

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Example 3.6.3.

Two dice are rolled. Are the events independent?

E : The roll is even.

2 : A 2 is rolled.

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Two dice are rolled. Are the events independent?

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2: A 2 is rolled.

No. Dependent.

The Product Rule

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$$p(A \cap B) = p(A | B) \cdot p(B)$$

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Example 3.6.4.

Two cards are dealt (without replacement). Find $p(J \cap H)$.

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Two cards are dealt (without replacement). Find $p(J \cap H)$.

J: A jack is dealt 1st

$$= p(H | J) \cdot \frac{4}{52}$$

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Example 3.6.4.

Two cards are dealt (without replacement). Find $p(J \cap H)$.

J: A jack is dealt 1st

H: A queen is dealt 2nd

$$\begin{aligned} &= p(H | J) \cdot \frac{4}{52} \\ &= \frac{4}{51} \cdot \frac{4}{52} = \frac{16}{51 \cdot 52} \end{aligned}$$

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Example 3.6.4.

Two cards are dealt (without replacement). Find $p(J \cap H)$.

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$$\begin{aligned} p(H \cap J) &= p(H | J) \cdot p(J) \\ &= p(H | J) \cdot \frac{4}{52} \\ &= \frac{4}{51} \cdot \frac{4}{52} = \frac{16}{51 \cdot 52} \\ &\approx .0060 \end{aligned}$$

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Example 3.6.5.

A cards is dealt, replaced, and another card is dealt. Find $p(J \cap H)$.

J: A jack is dealt 1st

H: A heart is dealt 2nd

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Example 3.6.5.

A cards is dealt, replaced, and another card is dealt. Find $p(J \cap H)$.

J: A jack is dealt 1st

H: A heart is dealt 2nd

$$p(H \cap J) = p(H | J) \cdot p(J)$$

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Example 3.6.5.

A cards is dealt, replaced, and another card is dealt. Find $p(J \cap H)$.

J: A jack is dealt 1st

H: A heart is dealt 2nd

$$\begin{aligned} p(H \cap J) &= p(H | J) \cdot p(J) \\ &= p(H | J) \cdot \frac{4}{52} \end{aligned}$$

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Product Rule for Independent Events

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A cards is dealt, replaced, and another card is dealt. Find $p(J \cap H)$.

J: A jack is dealt 1st

H: A heart is dealt 2nd

$$\begin{aligned} p(H \cap J) &= p(H | J) \cdot p(J) \\ &= p(H | J) \cdot \frac{4}{52} \\ &= \frac{4}{52} \cdot \frac{4}{52} = \frac{16}{52^2} \end{aligned}$$

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Example 3.6.6.

Two dice are rolled. Find $p(4 \cap 2)$.

4: The 1st die is 4

2: The 2nd die is 2

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Example 3.6.6.

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$$p(4 \cap 2) = p(4 | 2) \cdot p(2)$$

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Example 3.6.6.

Two dice are rolled. Find $p(4 \cap 2)$.

4: The 1st die is 4

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$$\begin{aligned} p(4 \cap 2) &= p(4 | 2) \cdot p(2) \\ &= p(4 | 2) \cdot \frac{1}{6} \end{aligned}$$

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Example 3.6.6.

Two dice are rolled. Find $p(4 \cap 2)$.

4: The 1st die is 4

2: The 2nd die is 2

$$\begin{aligned} p(4 \cap 2) &= p(4 | 2) \cdot p(2) \\ &= p(4 | 2) \cdot \frac{1}{6} \\ &= \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36} \end{aligned}$$

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Product Rule for Independent Events

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Example 3.6.7.

Two dice are rolled. Find $p(2 \cap E)$.

E: The roll is even.

2: A 2 is rolled.

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Example 3.6.7.

Two dice are rolled. Find $p(2 \cap E)$.

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Two dice are rolled. Find $p(2 \cap E)$.

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$$\begin{aligned} p(2 \cap E) &= p(2 | E) \cdot p(E) \\ &= p(2 | E) \cdot \frac{18}{36} \\ &= \frac{1}{18} \cdot \frac{18}{36} = \frac{1}{36} \end{aligned}$$

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Mutually Exclusive

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Recall:

Two sets, A and B , are *mutually disjoint/exclusive* if $A \cap B = \emptyset$

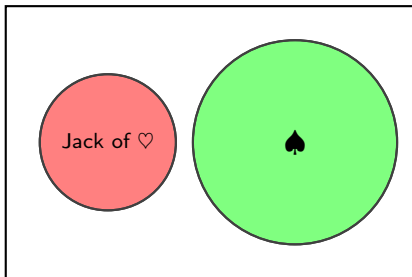
Two events, A and B , are *mutually exclusive* if $A \cap B = \emptyset$

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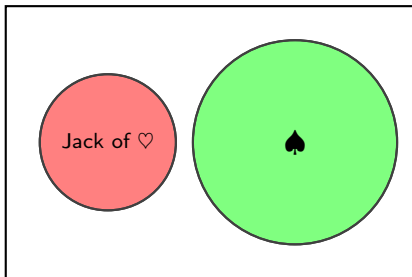
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- Recall: If $A \cap B = \emptyset$ then $p(A \cup B) = p(A) + p(B)$

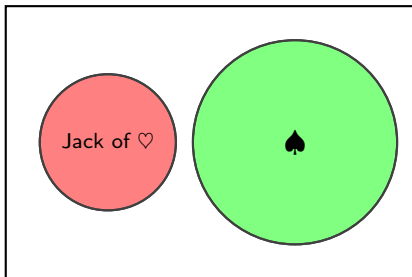


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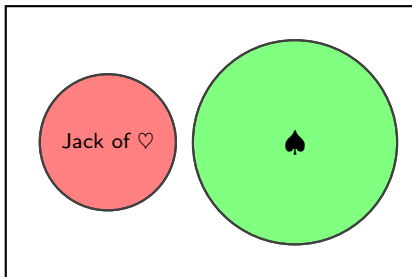
ex. The probability of drawing a $J♥$ OR a $♠$ is:

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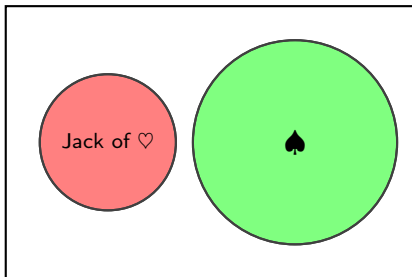
$$p(J♥ \cup ♠) = \frac{1}{52} + \frac{13}{52} = \frac{14}{52}$$

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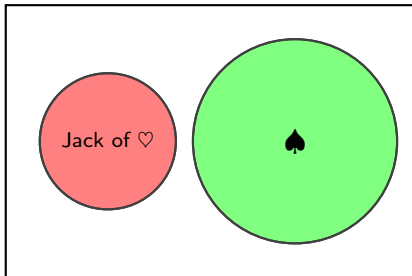
ex. The probability of drawing a $J♥$ AND a $♠$ is:

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- Recall:

ex. The probability of drawing a $J♥$ OR a $♠$ is:

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ex. The probability of drawing a $J♥$ AND a $♠$ is:

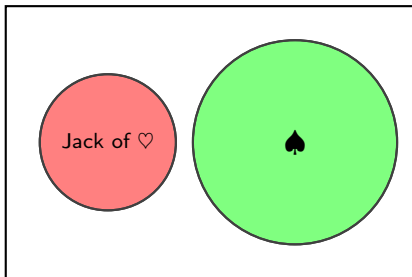
$$p(J♥ \cap ♠) = \frac{n(J♥ \cap ♠)}{n(S)} = \frac{0}{52} = 0$$

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Two events, A and B , are *mutually exclusive* if $A \cap B = \emptyset$



- Recall: If $A \cap B = \emptyset$ then $p(A \cap B) = 0$

ex. The probability of drawing a $J♥$ OR a $♠$ is:

$$p(J♥ \cup ♠) = \frac{1}{52} + \frac{13}{52} = \frac{14}{52}$$

ex. The probability of drawing a $J♥$ AND a $♠$ is:

$$p(J♥ \cap ♠) = \frac{n(J♥ \cap ♠)}{n(S)} = \frac{0}{52} = 0$$

Problem 3.6.1. A group of doctors found the following about their patients

Problem 3.6.2. A card is dealt from a full deck

- a. Find the probability of being deal a jack, $p(J)$.

Problems

- Exam 3 is on Monday 27th October, 2014

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- The review session is from 7pm-9pm on Sunday 26th October, 2014 in the RI classroom of the library.